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HIS notes

health records and information technology (Kenya Medical Training College)

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HEALTH INFORMATION SYSTEM

(HIS)

Mr. Mike njogu

Isiolo kmtc

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HEALTH INFORMATION SYSTEM

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UNIT 3	SOURCES OF HEALTH DATA	30 HOURS
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UNIT 1 INTRODUCTION TO HEALTH INFORMATION SYSTEMS (6 HOURS)

Define health information system

Health

It is a state of complete physical, mental and social well being and not merely the absence of disease or infirmity.

Information

A collection of facts that have been processed into usable form

System

- A collection of units that form a body
- A collection of components that work together to achieve the common objective of the whole.

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Define H.I.S

1. It is information about the people's health a how the community, government, NGO's uses the information to plan and provide quality health care services.
2. Organized and interpreted data about health status, resources, activities and events that relate to health.
3. A system that collects and presents health information and is responsible for the creation and maintenance of a data base.

Concept of health information system

Health information system is a system for the collection and processing of data from various sources and using the information for policy making and management of health services. It could be paper based or electronic. To apply this to nation, a county HIS integrates data from civil/vital registration, censuses, population surveys, facility surveys, individual records, services records and resource records for policy making and efficient management of services. There is a strong feeling that HIS is a key component of any health system. However, currently the data sources are incomplete and fragmented. The challenge is to bring together or integrate all these diverse data sources into one seamless system. Already there is good progress towards this integration by the use of DHIS

Health information management

It is the practice of maintenance and care of health records by traditional (paper) and electronic means in hospitals, physician's office clinics, health departments, health insurance companies and other facilities that provide health care or maintenance of health records. With the widespread computerized of health records and other information sources, including hospital administration functions and health human resources information, health informatics and health information technology are being increasingly utilized in information management practices in the health care sector.

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Historical background of Health Information system

In 1972, a committee including representatives from, Ministry of Health, Central Bureau of statistics and the Attorney General chamber was formed.

Its task was to design a health Information System for Kenya.

A pilot project was designed and tested in three Districts;

- Mombasa representing an urban District
- Kitui representing rural District
- Kwale representing peri-urban District

The pilot survey was completed in 1976 and its suggestions and recommendations were adopted.

One of the recommendations was to merge two units that is; Research and Evaluation Unit of the Division of Family Health and that of Vital Statistics Unit of the Division of Communicable Disease control into one entity of HIS.

In 1982, another committee of professionals was formed to investigate the accuracy and efficiency of all health data collection forms.

The committee suggested a number of changes and as a result all officially approved forms were changed from MED to MOH with a corresponding code number.

MED was a reflection of the Dept of Medical of one ministry of health (London) when Kenya was a British Colony.

In 1984, the National policy on District Focus for Rural Development was initiated. In response to the new development the ministry of health decentralized its reporting activities by establishing HIS offices in all Districts, where all data from all health facilities within the District are processed.

Goals and objectives of health information system

Goal of HIS

The ultimate goal is to develop an accessible and comprehensive HIS which is flexible to the needs of;

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- Patients
- Health care providers
- Health managers
- Evaluators and allied health personnel.

Objectives of health information system

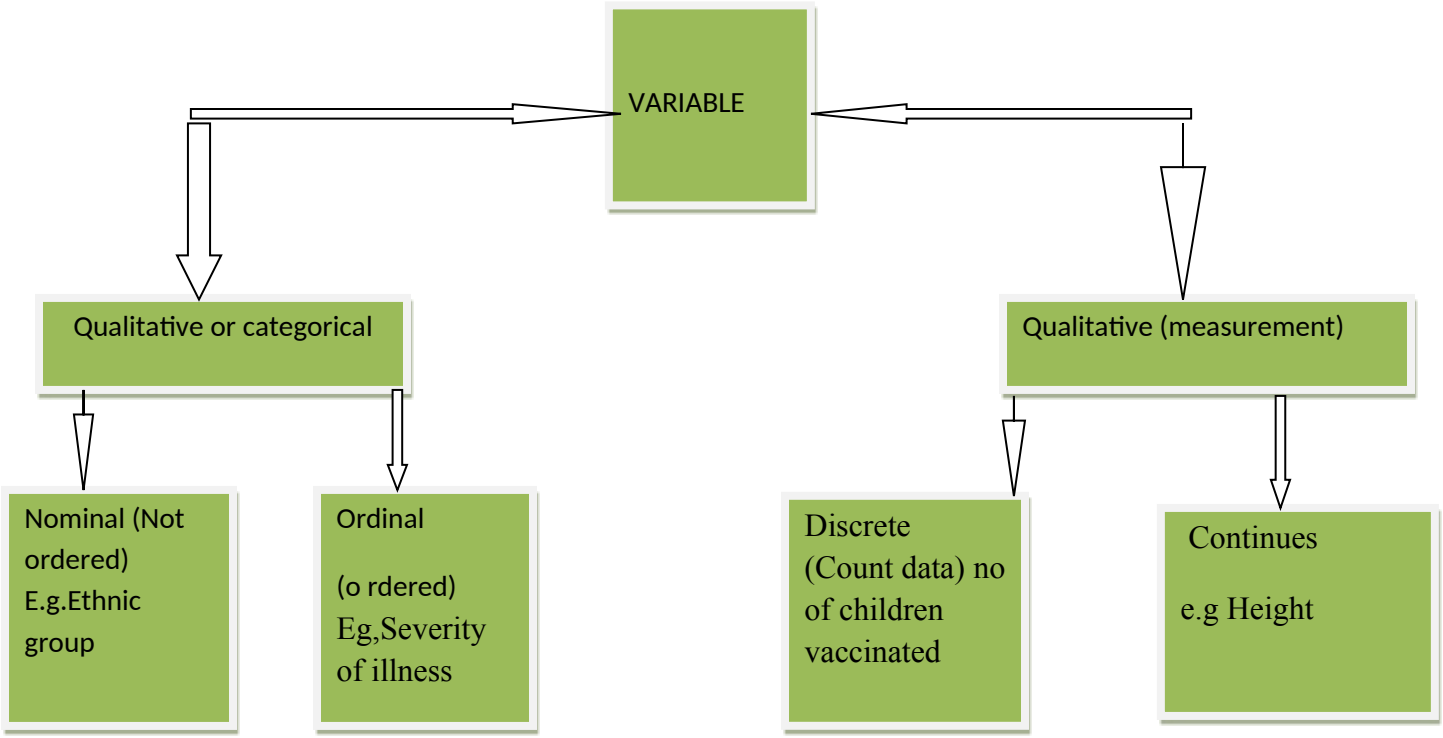
A good and effective HIS should;

1. Provide quantitative and qualitative data which is essential for identifying major programs.

NB; what is a variable?

It is any attribute or event that can have different values.

what is Qualitative and quantitative ?



2. Generate information to be used to evaluate health policies for planning health programs and efficient management of health services.

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3. Open up continuous dialogue between those who collect and the users of the information with the aim of improving health services.
4. Disseminates timely information to users' e.g. researchers and health administrators.
5. Provide regular feedback to all personnel in the information system which helps to emphasis the important role each of them plays.
6. Create and maintain a health information data base at the Sub County.

Personnel involved in HIS

The various individual involved in a typical HIS include;

- Families and communities
- Field health workers e.g Public Health Officers etc
- Clinical staffs e.g Doctors , Nurses etc
- National program health managers and planners
- Sub county health managers
- County health managers
- Health records and information officers
- All the other health workers

Different systems within the HIS

1. In patient health information system
2. Outpatient health information system
3. Management information system

1. Outpatient records

In the outpatient we have information on preventive, curative, health promotion and rehabilitative services. We have information on vertical programs such as

- Patient support centre
- KEPI currently DVI

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- Maternal and Child Health
- Family planning.

Each of these vertical programs has its own health information system. Each information system gives different health statistics for example from DVI we can get immunization coverage and the number of mothers who have received tetanus toxoid

The information can be used for planning and management of the health facility.

Several reports are prepared at the outpatient department such as;

2. **In patient records**

In the inpatient department we collect information on curative services from all patients stay in the health facilities. Upon discharge the health records personnel collects patients records and calculates the following hospital statistics, Bed occupancy, ALOS, TOI, VBD etc.

The statistics are used to monitor the operational efficiency of the health facility.

3. Management information system
    a) Personnel information system; gives the available manpower and how they have been deployed.
    b) Facility information system; gives the number of health facilities and the services they are offering e.g Family planning, immunization etc.
    c) Transport information system; gives the number of vehicles and those that are in working condition.

Standard format of health information system

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Health information policy

Uses of health information

Information is basically used in

1. Planning
2. Policy making
3. Decision making
4. Allocation of resources
5. Distribution of personnel
6. Administrative purposes
7. Research
8. Budget
9. Monitoring and evaluation
10. Forecasting
11. Health education
12. Intersectoral collaboration
13. Community participation

Functional HIS

Advantages of a functioning health information system:

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1. Information is readily available when requested or required, avoiding the last minute rushes to look for data.
2. Policy makers and managers of program use the information for the health and development planning and implementation and are sensitized on importance and effective utilization of Information.
3. Resource mobilization and allocation. Once data is readily available, it is easy to project the required resources by extrapolating from the past trends with a fairly good accuracy. Rule of the thumb should not be the norm or the practice.
4. Monitoring trends – Disease surveillance is functional and it is thus easy to predict the occurrence of epidemics of diseases.
5. Problem and gap identification – Various departments and institutions becomes part of the system; there is collaboration and co-ordination among various actors.
6. The work and duties of the staff that are involved in the system is appreciated and recognized.
7. Ad-hoc surveys and consultancies are avoided.
8. Decision making and priority service setting.
9. Operational research and training becomes easy and enables quick retrieval of the evidences in practice and use evidence based practices for demonstration.
10. Monitoring and evaluation of projects and programmes will be made easier and alongside working corrective measures or change of strategies will be realized in the initial stages.
11. Donor influence in Health information system is also avoided as in case of reviews.

Guiding principles:

The approach recommended for strengthening National Health Information System (NHIS) follows a number of principles which have been derived from the assessment of HIS development experiences around the world and WHO past efforts to support such development. These principles relate not only to information system design and use, but also to more effective styles of Technical co-operation.

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The overall purpose of these principles is to suggest that health information systems development must proceed as an integral part of efforts to strengthen the health care system and that **health data be recorded and used firstly, in support of individual care**, **local health service operation and community action**.

Placing emphasis on the use of data within the services will help increase the completeness and validity of the data that are selectively reported for supporting decision -making at all levels of the health system and for supporting health system, planning and development including service integration. Therefore:-

1. Strengthening the Health Information System at the various service levels should be undertaken in support of efforts to develop health services and improve their **performance.**
2. Any data to be recorded at any service level **MUST** have an explicitly identified use (for decision or action in terms of case or community management by staff or community members) at all levels. An implication of this principle is that no data should be requested from service levels to be reported to higher levels which do not have an actionable use at the recording level as well as at the receiving levels.
3. Any changes or developments to data recording and reporting should be made only to improve the provision of care at the patient and community level, particularly for those populations most in need. One implication of this principle is that countries, municipalities or regions should not be encouraged to change their information system primarily to provide data for central level and International reporting purposes.
4. Each Health administration must assess its needs for clinical and managerial information based on its mission, goals, priorities, core service responsibilities, levels and functions, models of service delivery, resources and access to information technology. The Health Information System should be designed so as to exceed the capability of the administration to manage it.
5. Great prudence should be applied when making changes to components of health information Systems that are working fairly well. This applies particularly to reporting systems of specialized services program such as family planning or Tuberculosis control or EPI. Overall revision in health service recording and reporting should rarely be

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undertaken. Changes in the system should aim to progressively integrate the recording and reporting system at the local service level.

6. Efforts should be made to make better use of existing data at all levels through practical analysis, improved presentation of data and efforts to improve the flow and sharing data across program and services.
7. Practical use of information should be encouraged and supported for database maintenance and report generation. Such computerization should normally employ generic software that is widely available database management systems and spreadsheets. Major computerization efforts should not dominate the selection of indicators and data or the design of records and reports, but support the improvement of data management through the application of these principles.
8. The selection and definition of a manageable data set or set of **"Essential health indicators"** is recommended as a sound activity for initiating the review and strengthening of health information systems and for devising a practical National Health Service monitoring capability. Core health indicators should be chosen for national, provincial and district use with the following criteria in mind:-

   (a) Useful for action. – The data needed for the indicator are useful for the person doing the recording with recorded data contributing to necessary action being taken with regard to the case, family, community or district being served.

   (b). Relevant for national and program monitoring.- the indicator can serve to measure progress toward stated national and program goals, objectives, targets, norms and standards. Such indicators focus on priority health problems in the country and the services and resources intended to manage those problems. Promotion and use of the indicator should strengthen routine program management and the health information system in the country.

   (c).Valid, consistent, reliable, representative and sensitive. - the indicators should possess the normal desirable characteristics of health data, example capable of being recorded across the services with the necessary degree of validity, consistency, reliability, representative of all population groups and be sensitive to short term changes in the variable of interest.

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(d) Ease of generation and measurement. - The indicator data should as much as possible result from normal service and surveillance, usually existing within routine records and reports.

(e) Understandable: - the indicator should deal with a single clear idea which everyone will see as an important measure. Composite indexes should be avoided.

(f) Ethical: - Data collection, including the choice of the data source, computation of the indicator and its use should not conflict with accepted ethical values.

UNIT 2 ORGANIZATION OF HEALTH SYSTEMS IN KENYA (4 HOURS)

Organization structure of Health Information System

Flow of health information system

Unit level

Sub county level

County level

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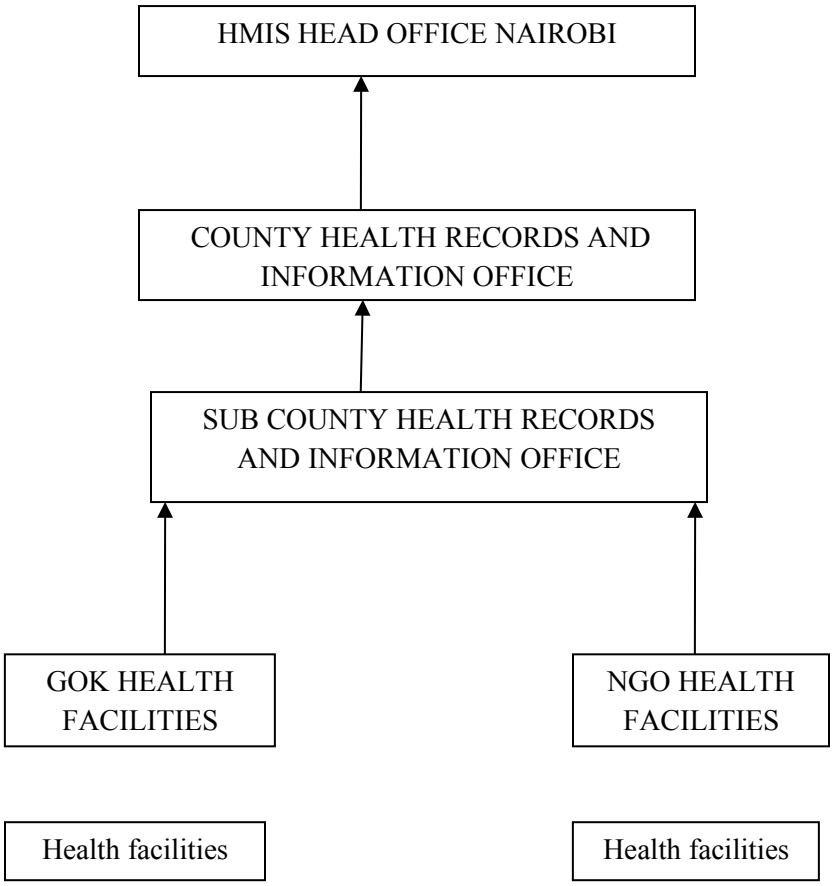
National level



The linkage Community unit levels of institution may Community unit ut in principle the flow of information is bottom top bottom. Data is collected, processed, analyzed, interpreted and a report written.

The principle that needs to be emphasized about health information collection and use is that data flows in systems and is made up of individuals and departments. If one link is weak, the entire system suffers.

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Data collection is closely related to data use. Any improvement in collection of data is not enough without a corresponding improvement in its use.

Managers who use data are likely to encourage its collection.

Health information is a vital resource for health planning, implementation and evaluation. Data users include everyone in the delivery system, ranging from individuals, patients, clients, county planners, national planners and international donor agencies.

Uses of data likewise range from monitoring of individual health, nutritional programs, evaluation and long term policy development.

It is absolutely essential to be selective and identify priority areas for data collection, since data collectors have limited capacity and data users have limited interests. The collection of data that will not be useful is wasteful.

Tiers of care

Community level

Dispensary and health center

District and sub District hospitals

Provincial hospitals

National referral hospital

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The Tiers of care according to KEPH (Kenya Essential Package for Health)

1. **Tier 1: Community Level** – This is a Community unit as defined in the previous KEPH Level 1 unit.

   What is a Community Health Unit?

   This is a health service delivery structure within a defined geographical area covering a population of approximately 5,000 people. Each unit is assigned 2 Community Health Extension Workers (CHEWs) and community health volunteers who offer promotive preventative and basic curative health services.

   It is governed by a Community Health Committee (CHC) and each Community Health unit is linked to a specific Health facility.

   WhyCommunityHealthUnits?

   To bring services closer to the people using innovative approaches that empowers them to be responsible for their own health.

Services Available at Community Health Units (CHUs)

a) Water and sanitation hygiene; e.g. Faecal management, Household water treatment and demonstrations on hand washing with soap, etc.

b) Advise on maternal and child health e.g. Immunization, Individual birth plan, etc.

c) Provision of Family planning commodities.

d) Growth monitoring for children under 5 years.

e) De worming of children.

f) Provision of Long Lasting Insecticide Treated Nets (LLITNs).

g) Management of diarrhea, injuries, wounds, jiggers and other minor illnesses.

h) Provision of Information, Education & Communication (IEC) materials

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i) Defaulter tracing (ART, TB and Immunization)

j) Referrals to health facilities

k) First Aid Services

2. **Tier 2: Primary Care Level** – Provision of basic outpatient health services – previous KEPH levels 2 and 3,dispensaries and health centers

   The network of ***health centers*** provides many of the ambulatory health services. Health centres generally offer preventive and curative services, mostly adapted to local needs.

   Dispensaries are meant to be the system's first line of contact with patients, but in some areas, health centres or even hospitals are effectively the first points of contact. Dispensaries provide wider coverage for preventive health measures, which is a primary goal of the health policy.

3. **Tier 3**: **County Level** – Provision of primary referral services – previous KEPH level 4

   District hospitals concentrate on the delivery of health care services and generate their own expenditure plans and budget requirements based on guidelines from headquarters through the provinces.

.

4. **Tier 4: National Level**: Provision of secondary and specialized services – previous KEPH Level's 5 and 6

   Provincial hospitals act as referral hospitals to their district hospitals. They also provide very specialized care. The provincial level acts as an intermediary between the national central level and the districts. They oversee the implementation of health policy at the district level, maintain quality standards, and coordinate and control all district health activities. Similar private hospitals at the provincial level include Aga Khan Hospitals in Kisumu and Mombasa

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National referral hospitals are at the apex of the health care system, providing sophisticated diagnostic, therapeutic, and rehabilitative services. The two national referral hospitals are Kenyatta National Hospital in Nairobi and Moi Referral and Teaching Hospital in Eldoret. The equivalent private referral hospitals are Nairobi Hospital and Aga Khan Hospital in Nairobi

UNIT 3 SOURCES OF HEALTH DATA (30 HOURS)

Sources of health information

1. Outpatient records
2. Inpatient records
3. Notification of infectious diseases
4. Registration of persons- (Births and deaths)
5. Surveys
6. Community based health care projects

Outpatient records

In the outpatient we have information on preventive, curative, health promotion and rehabilitative services. We have information on vertical programs such as

- Patient support centre
- KEPI currently DVI
- Maternal and Child Health
- Family planning.

Each of these vertical programs has its own health information system. Each information system gives different health statistics for example from DVI we can get immunization coverage and the number of mothers who have received tetanus toxoid.

The information can be used for planning and management of the health facility.

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Several reports are prepared at the outpatient department such as;

In patient records

In the inpatient department we collect information on curative services from all patients stay in the health facilities. Upon discharge the health records personnel collects patients records and calculates the following hospital statistics, Bed occupancy, ALOS, TOI, VBD etc.

The statistics are used to monitor the operational efficiency of the health facility.

Notification of infectious diseases

These are limited to infectious diseases. They are important indicators that show the presence of communicable diseases in the community. It helps to identify the need for control and prevention action.

Registration of births and deaths

The information on births and deaths is a vital element of vital statistics. Health begins at birth and end in death.

This information is important in projecting population growth rate and mortality rates.

Surveys

They are periodic methods of collecting health care data using sample population. There are two types;

- Formal surveys; they are sample based surveys. They are scientific in design.
- Informal surveys; they are like home visits conducted by health workers.

Community based health care projects

They are run by NGO's. Its main aim is to stimulate individuals and local communities to determine their own health care needs. They also initiate interventions.

In the CBHCP, people determine their own needs and decide on PHC activities to solve their health problems on a self help basis. Individuals, families and community members should be encouraged to participate in health promotion activities.

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Data collected from CBHCP may include information about;

1. Local people and their distribution
2. Diseases they suffer from
3. Organization of the local health services.

Uses of health information from CBHCP

Health information is one of the most essential components of an effective health delivery system. The uses include;

1. Parents use height and weight information to monitor the nutritional status of their children(BMI= weight in kgs/height in m^2)
2. Community groups use H.I to identify and correct environmental hazards.
3. Health workers use H.I to identify and screen persons who need treatment.
4. Clinical staffs use patients' history to diagnose and treat individual health problems.
5. Managers use H.I to maximize program effectiveness and efficiency and compare alternative methods of health care services.
6. Administrators use H.I to identify problems and audit the use of program funds and other resources.
7. Epidemic control units which use information for surveillance (predict outbreaks).
8. To help donor organizations identify priority areas to justify new program funding.
9. International agencies use H.I to monitor progress towards health goals (MDG's).
10. H.I is used for purposes of research.
11. To justify manpower training.
12. To help evaluate the impact of all health programs initiated.

National tools used in the collection of information in Kenya;

Common terminologies

Baseline:

1. Information gathered at the beginning of a study against which variations found in the study are measured.

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2. A known value or quantity with which an unknown is compared when measured or assessed

3. The initial time point in a clinical trial, just before a participant starts to receive the experimental treatment which is being tested. At this reference point, measurable values such as CD4 count are recorded. Safety and efficacy of a drug are often determined by monitoring changes from the baseline values.

4. A value representing a normal background level or an initial level of a measurable quantity and used for comparison with values representing response to an environmental stimulus or intervention.

5. A known value or quantity used to measure or assess an unknown, for example a baseline urine sample.

Baseline data

1. A set of data collected at the beginning of a study or before intervention has occurred.

2. **Baseline:** a reference point used to indicate the initial condition against which future measurements are compared.

Population: denoted (n)

1. The total number of people inhabiting a specific area.

2. The set of individuals, items, or data from which a statistical sample is taken.

3. All the organisms that constitute a specific group or occur in a specified habitat

4. All of the animals in a specifically defined area considered as a whole. The population may also be defined in modes other than geography, for example the human population.

Parent population: the original population about which it is hoped to make some inferences by examination of a sample of its constituent members.

Population proportion: the percentage of the population that has the subject characteristics.

Population pyramid: a graphic presentation of the composition of a population with the largest group forming the baseline, the smallest at the apex.

Population at risk: the individuals belonging to a certain group or community who have the potential to contract a medical condition or

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The population which is composed of animals that are exposed to the pathogenic agent under discussion and is inherently susceptible to it. High or special risk groups are those which have had more than average exposure to the pathogenic agent.

Population size: actual counting of a total population, the *census* method, is not often possible in large human populations. Alternatives are by various sampling techniques including *area trapping,* the trapping of all animals in an area, the *capture-release-recapture* method, the *nearest neighbor* and *line transect* methods,

The population size is expressed as the population present at a particular instant. Alternatively it can be expressed as a number of person-duration expression when the population is a shifting one and it is desired to express the population size over a period (e.g. persons per day).

Stable population: a population which has constant mortality and fertility rates, and no migration, therefore a fixed age distribution and constant growth rate.

Target population: in epidemiological terms the population from which an experimenter wishes to draw an unbiased sample and make inferences about it.

Catchment area: The surrounding area served by an institution, such as a hospital or school.

Catchment population;The population being served by an institution such as a hospital or school.

Eligible population; the number of persons targeted for a particular indicator.

Monitoring

Evaluation

Input; the resources need to achieve the results.eg. Finances, supplies,commodities, facilities etc.

Process; the activities you do to achieve results e.g providing new contraceptive methods, improving services, developing systems, conducting trainings etc.

Output; immediate results of activities e.g number of new family planning users, total number of users, number of people trained.

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Outcome short term change e.g change in service use comparing one period to another, change in knowledge, attitude or practice of a population or client group.

Impact the long term change among a population e.g changes in the contraceptive prevalence rate, total fertility rate, child morbidity and mortality rate.

NATIONAL ELIGIBLE POPULATION AND PROPORTIONS

POPULATION	PROPORTION
Under 1 year	4.5% of total population/catchment population
Under 5 years	18 % of total population/catchment population
(0-11) months	20 % of under 5 years/catchment population
(6-11) months	80 % of (0-11) months/catchment population
(9-11) months	25 % of (0-11) months/catchment population
(12-59) months	80% of (0-59) months/catchment population
(9-59) months	85 % of under 5 years or (9-11) months+(12-59) months
(5-14) years	29 % of total population/catchment population
Under 15 years	48 % of total population/catchment population
WRA(15-49) years	24 % of total population/catchment population
Elderly 60 years and above	5.2 % of total population/catchment population
Live births	Expected /estimated no. of deliveries
Expected pregnant women	4.5% of total population/catchment population
Surviving infants	Total live births- total infant deaths
Estimated no .of Emergency obstetric complications	0.75% of the expectant/pregnant women
Estimated no .of abortions	0.75% of the expectant/pregnant women

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NOTE: The above proportions are derived from national population of 2002 projected from the Kenya 1999 population census but the proportions varies from district to district.

Source: Analytical report on population projections Volume VII August 2002.

Data collection

It is the processes of gathering raw facts or figures. It provides the basic raw data from which further analysis is made and conclusions drawn. For any project or program to be acted upon it must be backed by variable data. Before collecting data obvious questions must be decided or answered.

1. Purpose of collection.
2. Scope of inquiry.(extent of the area/subject matter that is to be dealt with)
3. Source of data; primary or secondary
4. Method of collection.
5. Degree of accuracy desired.

Primary data: it is data that has been collected by the investigator for a certain objective and it's therefore original in character. E.g. population census collected by the Kenya National Bureau of statistics.

Secondary data; it is data which has been used in an investigation but originally collected by someone else. Therefore if someone else uses the KNBS data, it may constitute secondary data to him.

Therefore primary data in the hands of one person becomes secondary in the hands of another user. Data is usually in any of the two forms.

Methods of data collection

DATA COLLECTION TOOLS

INTRODUCTION

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The inability to generate reliable information needed to make decisions based on evidence is a major obstacle to public health services. Public health decision-making is critically dependent on the timely availability of sound data. The role of the Health Information System is to generate, analyze and disseminate such data. "**Nothing exists until it is measured"** the physicist Niels Bohr famously stated in 1930. The work of John Snow during the Cholera epidemics that ravaged London in the mid 1800s is an early example of an epidemiological investigation. Snow's groundbreaking work was made possible by the **registers of births and deaths maintained** by local authorities in every English parish from the early 1800s. Without information on numbers of deaths from cholera and **the street address of each victim,** Snow's **mapping of mortality** in relation to the sitting of water pumps would not have been possible.

Counting deaths is one component of the health information system, an integrated effort to collect, process, report and use health information and knowledge to influence policy-making, programme action and research.

The critical elements in Health Information System (HIS) are usually to turn raw data into useful form and extrapolate this through sector indicators. The main components are:-

- Data collection
- Data processing
- Data reporting
- Data dissemination and use.

Types of tools

Primary tools; Registers, forms

INTERGRATED MINISTRY OF HEALTH DATA COLLECTION AND REPORTING TOOLS

DATA COLLECTION TOOLS (REGISTERS)

1. MOH 204 A OUTPATIENT REGISTER (under five years)

2. MOH 204 B OUT PATIENT REGISTER (over 5 years)

3. MOH 511 CHILD WELFARE CLINIC (CWC) REGISTER

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4. MOH 510 IMMUNISATION REGISTER FOR CHILDREN

5. MOH 333 MATERNITY (DELIVERY) REGISTER

6. MOH 406 POSTNATAL REGISTER

7. MOH 512 DAILY ACTIVITY (FAMILY PLANNING) REGISTERS

8. MOH 301 IN-PATIENT REGISTER

9. MOH 209 RADIOLOGY REGISTER

10. MOH 240 LABORATORY REGISTER

11. MOH 268 DIAGNOSTIC INDEX CARD

12. MOH 405 ANTENATAL CLINIC (ANC) REGISTER

13. MOH 361A Pre-ART Register

14. MOH 361B ART Register

15. MOH 366 Care and treatment activity register

16 MOH 257 COMPREHENSIVE CARE CLINIC PATIENT CARD (Blue card)

MATERNAL AND CHILD HEALTH BOOKLET MOH 216

The Maternal and Child Health Booklet is a revised version and combination of the Antenatal card and Child Welfare Card. The first part contains the mother's full antenatal and post natal profile.

(i) ANC

(ii) Postnatal

(iii) Second part contains the child's details on immunizations and other services delivered to a child before age 5 years that will be detached from the booklet.

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In case of multiple deliveries the health worker should initiate a booklet for each child.

The health worker is advised to be extra careful while filling the information in the booklet by ensuring that correct information is recorded in the appropriate spaces provided. The growth monitoring charts should be marked progressively as the child grows. Health workers are advised to share the information pertaining to the child with the mother or care taker.

INSTRUCTIONS ON DATA REPORTING TOOLS (Summary forms) Secondary tools;

Sample tally sheets and summary sheets are appended below:

1. MOH 105	Service delivery
2. MOH 701 A	Under five (<5) years Daily outpatient morbidity tally sheet.
3. MOH 701 B	Over five (>5) years Daily outpatient morbidity tally sheet.
4. MOH 705 A	Under five (<5) years Daily outpatient morbidity summary sheet.
5. MOH 705 B	Over five (>5) years Daily outpatient morbidity summary sheet.
6. MOH 702	Immunisation and Vitamin "A" Tally sheet
7. MOH 710	Immunisation and Vitamin "A" summary sheet.
8. MOH 704	Child Health and Nutrition Information System tally sheet.
9. MOH 711 A	Integrated tool for RH, HIV/AIDS, Malaria, TB, and Child nutrition health facility summary.
10. MOH 711 B	Integrated tool for RH, HIV/AIDS, Malaria, TB, and Child nutrition District summary.
11. MOH 717	Monthly Workload report for hospitals (Service workload for all areas)
12. MOH 268	Diagnostic Disease Index
13. MOH 718	In-patient morbidity and mortality summary sheet.
14. MOH 708	Environmental Health services

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15. MOH 715 Semi Annual Health Facility services inventory form

16. MOH 505 weekly epidemic monitoring form

17. MOH 515 community health extension worker summary

18. MOH 731 Comprehensive HIV/AIDS facility reporting form

17. MOH 216 maternal and child health booklet

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GENERAL GUIDELINES FOR FILLING THE SUMMARY FORMS:

(Procedures of completing reporting tools)

1. The first step in completing data reporting forms is to ensure that all the identification particulars are filled in before completing the particular form. These are the names of the province, district, constituency, facility and the period for which the report is covering.

2. Specific ages should be reported within the appropriate age classification.

3. In forms where data is disaggregated by sex, the appropriate data should be filled in the correct column or spaces provided.

4. Care must be taken to separate new or first visits and re-visits or re-attendances.

.**Note: New or first visit** –these are patients/clients who come to your facility for the First time. While Revisit n**or Re-attendance-** these are patients/clients who make subsequent return visits after the first visit.

5. Where there is a provision to show totals, they **MUST** be aggregated.

6. While making entries in the forms, accuracy **MUST** be maintained to avoid errors or transposition of figures.

7. While reporting, completeness **MUST** be observed. No spaces should be left blank and no dashes. You are instead encouraged to practice zero reporting.

8. The name of person preparing the report, the date the report is being completed and the commitment signature **MUST** be filled in.

9. Once the reports are completed, they are supposed to leave your facility before the 5th of the following month to the District Health Information System (DHIS).

10. At the district, once all the reports from the health facilities have been received, summaries should be promptly made.

11. The district **MUST** maintain a checklist of all reports and all facilities and check the reports against the facilities to ensure completeness and timeliness.

12. Using a copy of the summaries made, districts **MUST** analyze and share the information at their level.

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13. Districts should submit the summaries to the province or national level **on or before 15th** of the following month and give feedback to the health facilities.

14. The province collects all the district reports, make copies (manual or electronic) do the analysis and use the information as they make arrangement to transmit the summaries to the national level **before 21st of the following month if data flows through the province inform of hard copies**.

15. Likewise, provinces should make and maintain a checklist of the reports and districts to ensure timeliness and completeness. They must give feedback to the districts and share the report at that level.

16. To maintain accuracy in recording data collected through tally sheets, health workers should tally from the registers on daily basis.

17. The national level data repository (HIS) should acknowledge receipt, process and analyze the data and give feedback to the lower levels and share the information horizontally and vertically.

18. All levels are encouraged to prepare annual reports that will encompass all activities, outputs and in-puts.

19. For communicable diseases that are for immediate reporting, such should be reported without further delay using the appropriate tools and channels for example using case based investigation forms.

How to handle the tally sheets

Tally sheets are working sheets on which data is recorded to facilitate ease of count at the time of making summaries.

Proper understanding of the content of each tally sheet is essential. The proper way of making a tally is to slash a zero with forward slash (**Killing one zero at ago)** for example 0

Tallies are normally made immediately a clinician is through with a patient/ client before attending to the next and at the end of the day or early next morning from the register. This depends on circumstances at the facility.

How to handle the Summary sheets

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Summary sheets will be completed at the end of the specified period, either from tally sheet or registers or any other source documents. As the health facility submits the summaries to the next level, a copy **must** remain in the health facility.

Summaries:

If the health information system forms are so complicated that those who are to complete them cannot understand them or the forms are not available all the time, data will not be collected properly. Therefore, every effort must be made to simplify forms and to ensure that there is a constant supply of them. Such forms can have a space for some simple on the spot analysis e.g. total numbers or percentage to be calculated.

Practical session in completing some of the tools below:-

- Weekly notification of infectious diseases
- Monthly notification of infectious diseases
- Outpatient morbidity report
- Inpatient morbidity and mortality report forms
- District outpatient morbidity summary forms
- Integrated RH, HIV, Malaria, TB and Nutrition
- Vaccine and immunization
- Service workload
- Nutrition monthly reporting
- HIV reporting tools

Community based health information system- CBHIS

- Community referral form MOH 100
- House hold register MOH 513
- Service delivery log book MOH 514
- Community health extension worker summary MOH 515

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UNIT 4 HEALTH SECTOR INDICATORS (20 HOURS)

Define indicators

- Indicators are statistical values which give insight into the quality of health services or health status.
- A Measurable marker of a change over time
- A Unit of Information Measured overtime that documents change in a specific condition.

Obviously indicators are intimately related to activities. Each issue or problem is measured in terms of indicators which define the magnitude of the problem.

Developing indicators is the 1st step in performing a meaningful analysis of Health Information.

The most useful method in developing indicators for health information is a review of the objective of the various programs of health services. If program objectives are clearly stated, developing indicators become a relatively simple task.

USES OF INDICATORS

- Setting performance targets and assessing progress towards achieving them
- Identify problem via early warning system to allow corrective action to be taken
- Indicating whether an in-depth evaluation or review is needed
- Benchmarking comparisons between different organizational units, sub counties/counties/facilities over time
- Indicators are tools used to convert raw data into useful information and to enable comparison between different programs or facilities.

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- They are observable Markers of progress towards defined targets
- Are used to describe the situation and to measure changes overtime
- Provide information about a broad range of conditions through a single measure (e.g. ALOS)
- Provide a yardstick whereby institutions or teams can compare themselves to others doing similar work
- Indicators allow managers to compare actual program implementation and results to work plans and predictions

To understand the operationalisation of the indicator, the following questions are essential:

- At what frequency should the numerator and denominator be collected?
- At what frequency should the indicator be processed and analyzed?
- Who will actually make use of indicators?
- What is the target (objective) of the indicator that needs to be achieved?
- What is the threshold .the minimum or the maximum
- What will be the nature of the action (decision) once the indicator reaches the threshold?

TYPES OF INDICATORS (Domains inputs, output, impact)

Input indicator

Input measures resources invested in the program in the program at the beginning of a phase or those that need to be available for implementation of program e.g. Drugs, vaccines, staffs, training, finances.

Process indicators

It measures the activities that must be performed or action normally done under several rules/standards e.g. procedure for treatment or making a diagnosis.

Output indicator

Measures the amount of activities completed or services/ goods delivered to target population e.g. number of children vaccinated, number of women bringing children for vaccination etc.

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Outcome/ impact indicators

These measure long-term eventual outcome /output of a program or changes in behavior or practice upon a population as a result of the program eg impact of program on morbidity and mortality levels compared to baseline.

Other Classifications of Indicators:-

- Morbidity and Mortality
- Morbidity Indicators include

Units of measurements; Rate, Ratio, Numbers,

Type of measurement (Calculating Matrix)	Description	Example
Count indicator	No. of events without a denominator	No. of new ARV clients
Proportion indicator	Numerator is contained in the denominator	35% of facilities do not have electricity
Rate indicator	Frequency of events in a specified time in a given population	Incidence of STI cases in a given population in a given year
Ration indicator	Numerator is not contained in the denominator	Ratio of patience to a doctor

Indicator

1. Count Indicators e.g. Number of Malaria cases
2. Proportion Indicators e.g. Percentage of WRA receiving FP commodities
3. Rate Indicators e.g. Facility Infant Mortality Rate

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4. Ratio Indicator e.g. Doctor Patient Ratio

QUALITIES OF A GOOD INDICATOR

1. SIGNIFICANCE- the indicator should represent information that is

- Important
- Useful for action
- Interesting.

2. VALIDITY –It should measure what is intended. Does the indicator represent information that can be correctly identified?

Measure what is intended.

3. RELIABILITY-Does the indicator represents the information that can be correctly quantified repeatedly.

Measurement of the indicator should be the same no matter how many times it is carried out.

4. **FEASIBILITY** (Possible) does the indicator represent information that can be collected with available resources?

5. **PRECISE DEFINATION-**Indicators should be defined using clear, specific terms

6**. TIMELINESS-**Indicators should be measured at appropriate time interval according to availability of data

7**. COMPAREABLE** -Indicators should be comparable across different population groups

8. **RELEVANT FOR NATIONAL AND PROGRAM MONITORING-**The indicator can serve to measure progress toward stated National and Program Goals, objectives targets, norms and standards. Such indicators focus on priority Health problems in the country and the service and resources' intended to manage those problems.

9. **EASE OF GENERATION AND MEASUREMENT**-The indicator data should as much as possible result from normal service and surveillance usually existing within routine records and reports

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10. **ETHICAL** –Data collection including the choice of the data source, computerization of the indicator and its use should not conflict with accepted ethical values.

Purpose

Frequency

Difference between coverage and performance

The analysis of data collected in the health information system is simple and straightforward. Not having a calculator is an invalid excuse for lack of analysis as analysis can be done without one. The most important data analysis is to estimate coverage for the services offered e.g. what proportion of children less than 1 year complete immunization schedule before their first birth day? Or what percentage of women delivers without attending antenatal clinics?

Percentage coverage = number of cases in clinic X 100

Total number of cases in catchment population

It is vital in data analysis to identify appropriate denominators and numerators. Population data are available from the past census and population projections.

Having estimated the catchment population, the next step is to estimate the target population for various services which is the denominator of the coverage fraction. E.g. catchment population = 25,000.

a) Antenatal care

Target population 4.5 Percentage of 25,000 = 4.5/100 * 25,000 = 1, 125

If number of new visits are 976 therefore coverage = 976/1125 * 100 = **86.76Percentage**

If number of re-visits are 2, 090 the mean number of visits per new attendant for antenatal care is new visits +

revisits divide by new visits.

= (976+2090)/ 976 = 3.1 visits Per new antenatal attendant.

b) Morbidity data

Morbidity data are collected in both outpatient and in-patient by age, and diagnosis. Considering new cases gives a picture of the morbidity pattern at the health facility for example;

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Data source; numerators and denominators

Application level sector, program, national, county, facility, community, MDGs

DATA QUALITY

Data Quality - the degree of excellence exhibited by the data in relation to the portrayal of the actual scenario.

-The state of completeness, validity, consistency, timeliness, and accuracy that makes data appropriate for a specific use.

Data Quality Assessment (DQA) - a process of verifying the quality of data, assessing the system that produces that data, and developing actions to improve both the process and the system

Routine Data Quality Assessment (RDQA) - A continuous/regular process of verifying the quality of data, assessing the system that produces that data and developing action plans for improvement

Quality Assurance (QA) - A process that focuses mainly on measuring compliance with established standards

Quality Improvement (QI) - A systematic process to improve the quality of health care by monitoring quality, finding out what is not working, and fixing the problems of health care delivery

Quality control- the process of monitoring and maintaining the reliability, accuracy and completeness of data

Dimensions/attributes of Data Quality

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- **Accuracy** - data that measure what they are intended to measure . its also known as validity
- **Reliability** - data that are generated based on protocols and procedures that do not change according to who is using them and when or how they are used
- **Timeliness** - data that are produced on time, adhering to standard schedules
- **Completeness** – the capture of all of the data and not just a fraction of what should be included
- **Confidentiality -** clients are assured that their data will be maintained according to national and/or international ethical standards; personal data are not disclosed
- **Precision -** all variables required for data analysis and use are captured, and the data have sufficient detail
- **Integrity -** the system used to generate data is protected from deliberate bias or manipulation

Threats to data quality

Impediments to good data quality

Three sources that threaten data quality

I. Inappropriate data collection instruments
II. Poor recording and reporting
III. Errors in processing data

Inappropriate data collection instruments and procedures.

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One important requirement of a well planned and executed information system is the development of a set of concepts to be covered and adherence of all stages of data collection and processing operations.

These concepts provide the basis for development of questions, wordings, instruments for data collectors and specifications, for editing, coding and tabulating data.

Imprecise definition of terms, lack of standardization, inaccurate translations of materials adopted from other context, leaving inadequate space for writing responses on data collection forms are some of the ways that can bring about errors.

Sometimes a well designed instrument might be used in an inappropriate setting for example, a sentinel surveillance system designed to classify the mechanism of HIV transmission in Sub Saharan Africa cannot be replicated in Western Asia without significant modifications of the data instrument.

The poor layout and quality of printed forms often contribute to low quality data in developing countries where resources are scarce. Forms are sometimes illegible due to quality of the original form and repeated photocopying.

Threats to data quality from Poor recording and reporting

(a) Failure to report data

Despite procedures for timely transmission of data between different levels of a health care system, sometimes data transmission does not occur.

Given the high cost of stationery (forms, operational cards, registers etc) and the inefficient supply system in many developing countries, the reporting health facility may lack access to necessary supplies or data transmission may be hampered by the frequent absence of personnel in charge of the information system or even if the personnel are available the high opportunity cost of time spent recording data on multiple forms coupled with lack of incentives for reporting may hamper regular reporting.

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Another problem may be lack of transportation and other means of communication (Electronic Data transfer).

(b) Unintentional recording and reporting of inaccurate data

Errors are sometimes introduced into the data for innocent reasons. Staffs may not have appropriate skills or supporting equipment to effectively carry out their data reporting responsibilities for example it is not uncommon in developing countries for the same reporting forms to be used at different levels of the health care system.

With computerized data entry a clerk with limited experience may enter codes in a data base by selecting from menu options that may not match the exact terminology used by the physician.

(c) Potential sources of Errors in recording clinical data

- Misinformation from the patients
- Quality of clinical findings
- Errors in recording patients data by providers
- Personnel errors during the recording of diagnostic tests
- Inaccurate interpretation of tests/results

(d) Intentional recording and reporting of false errors

One of the greatest fear of users is that information system will monitor employees work and that repercussions will occur if the employees do not attain a particular level of performance indicator.

Sometimes the health system is set up in such a way that reporting accurate data acts as a disincentive.

Physicians at times deliberatively mis-record patient data to avoid litigation.

Errors in processing data

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Errors occur during the transfer and aggregation of data at all levels of the information system. Manual processing which is common in most developing countries involve tedious computation that often result in errors. Corrections are rarely made for missing reports.

If computers are available, data processing operations typically involve a serious of basic steps each of which is suitable to possible introduction of errors;

- Editing
- Coding
- Data entry
- Tabulation

(a) Editing

Registers and forms are edited to correct inconsistencies and eliminate omissions. During this stage managers need to make decisions about treatment of "unknown" responses in the raw data.

Where information is lacking on one variable, can a reasonable entry be estimated based on other information supplied in the form?

(b) Coding

It is the conversion of entries on forms or registers into symbols. While many coding procedures are relatively simple, involving classification of individuals by sex (male, female), by location (village x, village y), reason for risk (preventive, curative) and so on, some coding procedures are complicated and have the potential for significant errors.

For instance, when classifying causes of death according to International Statistical classification of Diseases 10th Revision (ICD 10) categories, it is relatively easy to confuse the coding of disease A37.0 (whooping cough due to bordetella pertusis) with that of category A37.1 (whooping cough due to Bordetella parapertusis)

(c) Data entry

Data entry refers to the means of transfer of the data from the original document to the computer. For manual data entry into the computer, many institutions perform double entry to reduce errors. With double entry a computer program accepts the record only if the clerk has entered it

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exactly the same way. This procedure minimizes the introduction of errors due to typing mistakes.

(d)Tabulation

A final step entails re-arranging the data in order to provide the most useful charts and figures for decision makers. The potential for errors is significant if the programmer provides incorrect instructions (to the computer) for the required tabulation. For instance a program manager might request for information on the number of ANC visits in District X (a rural accessible area) in 2000.

The programmer might incorrectly insert the code for District Y (an urban area of high access) for 2001.

Without reviewing the coding from the programming, the program manager might recognize that District X and Y were confused, but unlikely to know that the years 2000 and 2001 were also tabulated erroneously.

What can be done to ensure data quality

The following strategies can help to ensure high quality data at low cost.

(a) Keep the design of the information system as simple as possible

Operational cards, registers and forms should be designed in a simple manner with clear instructions. Valid, sensitive and specific instructions MUST be selected to increase data quality. It is also important to minimize the number of levels in the information system in orders to avoid errors during the transfer and processing of data.

In general the collector and the user of data should be as close as possible. Indeed, in action led information systems, the collectors and users of data may be the same people. This will reduce errors in handling data and will improve decision making process as data is more likely to be meaningful to the person who collects them.

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(b) Involve users in the design of the system

Involving users in the design of the system is an incentive in two ways:-

i. It helps develop an information system that is best suited to the user
ii. It increases ownership of the system

While involving users takes more time, it promotes a general understanding of the system and improves the quality of the data produced. Furthermore it ensures that the data collected is relevant.

(c) Design of the data collection instruments

Prevalence rate

Incidence rate

Mortality rate

Morbidity rate

Fertility rates

The various measures that can be used to express fertility levels in a given community:-

- Child woman ratio
- Crude death rate
- Age specific fertility rate
- Total fertility rate.

(a) **Child woman ratio**

Population aged (0-14) / population of women of CBA (15-49)

(b) **Crude birth rate**

It is the most basic form of fertility measurement which is a ratio of birth occurring in the year to the population of a given geographical area multiplied by 1000 as a common expression

$$CBR = \frac{\text{No of births during the year}}{\text{Population during the year}} \times 1000$$

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The resultant answer is expressed as_____births per 1000 population

(c) **General fertility rate**

This is a measure of fertility that seeks to overcome the influence of SEX and AGE composition on the population in the presentation of fertility rate or measurement. It shows the number of births per 1000 women of child bearing age.

$$GFR = \frac{\text{No of births during the year}}{\text{Population of women aged (15-49)}} \times 1000$$

The resultant answer is expressed as_____ births per 1000 women of CBA

Example: A population distribution had 175,973 births and a female population of 297,533 women aged 15-49 years.

Calculate GFR

$$= \frac{175{,}973}{297{,}533} \times 1000 = 591.44 = 592$$

The GFR for the area is 592 births per 1000 women of (15-49) years

(d) Age specific fertility rate

$$ASFR = \frac{\text{No of births in a specific age}}{\text{Population of women in a specific age}} \times 1000$$

	1989			**1999**		
Age	No of women	No of Births	ASFR	No of	No of Births	ASFR

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				women		
15-19	1,200,712	204,054	170	1,721,194	273,738	160
20-24	1,013,340	172,211	170	1,504,389	239,258	160
25-29	847,287	143,992	170	1,164,594	185,218	160
30-34	575,651	97,829	170	845,230	134,425	160
35-39	457,942	77,825	170	723,749	115,105	160
40-44	364,244	61,902	170	516,989	82,222	160
45-49	293,405	49,863	170	418,987	66,635	160
	4,752,581	**807,676**		**6,895,132**	**1,096,598**	**1120**

(e) Total fertility rate (TFR)

It is considered the best measure of fertility. Also referred as to as the TOTAL FERTILITY PERIODIC RATE

- No of births per specific age group
- Total no of women in the specific age group
- Add all the ASFR

Substitution:-

$$TFR = \frac{\text{sum of ASFR}}{1000} \times 5$$

If every woman lives up to the age of 49 years then each would have given birth to 5 children

Mortality measures

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mortality is obtained from the noun “mortal” which refers to loss of life or generally it pertains to death. h

UNIT 5 USE OF ICT IN HEALTH INFORMATION SYSTEM (30 HOURS)

Data handling

ICT application

<u>DHIS (Theory and practical sessions)</u>

- DHIS 2 stands for District Health Information Software – Version 2
- Originated from University of Oslo, Norway, where it was developed under a global research and development initiative
- Distributed freely - is Free and Open Source Software (FOSS)
- DHIS 1 developed in 1997
- Based on comments and feedback from the field level use, the University of Oslo initiated the process of developing the second version of DHIS (DHIS 2) in 2005
- Currently is the national standard in South Africa and is used in all the health facilities in the country.
- DHIS 2 is being used in many countries in Africa such as Ethiopia, Nigeria, Botswana, Tanzania, Zambia, and various other countries

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Requirements

Computer (or Mobile Phone) and Internet Connectivity

Browser (Google chrome, Mozilla 8, Internet Explorer 9 and above

Accessibility to Internet

Website address:-

- Live site- www.hiskenya.org
- Training Site-http://178.79.169.90/

Log in Credentials

Details Required to Create User & Password

- ☐ Your name (First name, Surname)
- ☐ E-Mail address
- ☐ Phone Number
- ☐ Your working location
- ☐ Job Title
- ☐ Why required to access to DHIS2

Design of DHIS

All data in one place – a central repository

- Supports collection of many different data sets (and can easily expand to new data requirements)
- Automated import from other systems like MFL, EMR, HR, Logistics etc. (planned for)
- Avoid duplicate systems for data collection and storage

Easy access to data to all online users

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- Wherever you are, as long as you have Internet and a username you can access the latest data
- DHIS 2 becomes a data centre for all stakeholders with access

Pre-defined analysis tools shared by all users

- both one click report and custom data analysis
- targeting different needs and skills
- User-defined dashboards to give a quick overview of the latest data
- GIS interface to all the data

Data aggregation to any level or period

- While data is collected and stored monthly by health facility, a report can show quarterly or yearly values by any level in the hierarchy

Built in support for definition and calculation of indicators (coverage, incidence rates etc.)

- Often referred to as "calculated indicators"
- Makes use of population and other denominator data already in the system

All reports and analysis tools support indicators and allow for analysis and comparisons across areas and over time

- Instead of looking only at raw numbers or doing manual calculations in Excel

Many output formats are supported; html, pdf, excel, jasper reports

DHIS Browsers

The DHIS 2 is a web-based application and is available in an Internet browser when you have Internet connection.

The recommended browsers are:-

- Google Chrome
- Mozilla 8
- Internet Explorer 9 and above

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Principles of DHIS Reporting

There two types of reports:-

- Standardized -One-click
- Dynamic-enhanced deep analysis

- Facilitates users with different levels of experience
- Facilitates requirements for fixed reports as well as deep analysis

Online system enables horizontal comparison

Feedback reports are available immediately

Possible to compare performance across organization unit

Types of Reports

(a) Standardized

- Standard Report
- Dataset Report
- Reporting Rate Summaries
- Report table
- Resources

(b) Dynamic

Pivot Table

Organization Unit Distribution Report

Tally sheet

DHIS in the Kenyan context

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DHIS was introduced in Kenya in 2010 and piloted in Coast in January 2011

National Roll out was from May-October 2011

Rationale for DHIS adoption in Kenya

- Multiplicity of HIS/M&E systems
- Inability to authoritatively state level of sector performance
- Difficulty determining intra-district reporting rates
- Limited analysis with often conflicting statistics
- Limited capacity of excel based databases
- Overall erosion of trust in the existing system

DHIS2 supports different facets of the information cycle including:

- Collecting data.
- Running quality checks.
- Data access at multiple levels.
- Reporting.
- Making graphs and maps and other forms of analysis.
- Enabling comparison across time (for example, previous months) and space (for example, across facilities and districts).
- See trends (displaying data in time series to see their min and max levels).

Benefits of DHIS

1. Provide a comprehensive HIS solution based on data warehousing principles and a modular structure which can easily be customized to the different needs of the health systems - and supports the idea of an integrated HIS at all levels of the health hierarchy.
2. Customization and local adaptation through the user interface. No programming required to start using DHIS 2 in a new setting (country, region, district etc.).
3. Provide data entry tools which can either be in the form of standard lists or tables, or can be customized to replicate paper forms.

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4. Provide different kinds of tools for data validation and improvement of data quality.
5. Provide easy to use - one-click reports with charts and tables for selected indicators or summary reports using the design of the data collection tools. Integration with popular external report design tools like iReport and BIRT allows super-users to flexibly add more custom reports accessible to all users.
6. Flexible and dynamic (on-the-fly) data analysis in the Data Visualize and the GIS modules.
7. A user-specific dashboard for quick access to the relevant monitoring and evaluation tools including indicator charts and links to favorite reports, maps and other key resources in the system.
8. Easy to use user-interfaces for metadata management e.g. for adding/editing datasets or health facilities. No programming needed to set up the system in a new setting.
9. Functionality to design and modify calculated indicator formulas.
10. User management module for passwords, security, and fine-grained access control (user roles).
11. Messages can be sent to system users for feedback and notifications. Messages can also be delivered to email and SMS.
12. Users can share and discuss their data in charts and reports using Interpretations, enabling an active information-driven user community.
13. Functionalities of export-import of data and metadata, supporting synchronisation of offline installations as well as interoperability with other applications.
14. Integration with other software systems – using the DHIS 2 Web-API and the Integration Engine.
15. Further modules can be developed and integrated as per user needs, either as part of the DHIS 2 portal user interface or a more loosely-coupled external application interacting through the DHIS 2 Web-API.

Practical session recommended

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MASTER FACILITY LIST

The MFL is the official list of all the facilities operating in Kenya, and is maintained by the SCHMT through the SCHRIO.

- The MFL is found on www.ehealth.or.ke/facilitylogin
- The MFL has a unique code for each facility
- The MFL contains permanent and semi-permanent bio-data on each facility:
- Permanent data is that which does not change – name, directions to locate the facility, geo-codes, etc

The MFL has two official sites:-

1. **Log-in site**- This site requires a username and a password to get in. It is only authorized personnel who can get into the Login site.
2. **Public Site**- This available for everyone to view and does not require to Login

Explain functions of the different Menus of MFL

Home Menu: - Provides a quick and faster method for searching of facilities and contacts.

Facilities Menu: - Provides for advanced filter based facility and contact searching, editing and adding of new facility to the MFL

Admin Offices:-This menu item is used to provide all facility contacts

Reports: - This menu item gives summarized reports about facilities

System: - The system menu item is primarily used for system configuration and administration

Master Community Unit List (MCUL) Practical session recommended

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Other applications of ICT

ELECTRONIC MEDICAL RECORDS

INTRODUCTION

Paper-based records have been in existence for centuries and their gradual replacement by computer-based records has been slowly underway for over twenty years in western healthcare systems. Computerized information systems have not achieved the same degree of penetration in healthcare as that seen in other sectors such as finance, transport and the manufacturing and retail industries. Further, deployment has varied greatly from country to country and from speciality to specialty and in many cases has revolved around local systems designed for local use. National penetration of EMRs may have reached over 90% in primary care practices in Norway, Sweden and Denmark (2003), but has been limited to 17% of physician office practices in the USA (2001-2003). Those EMR systems that have been implemented however have been used mainly for administrative rather than clinical purposes.

- Electronic medical record systems lie at the centre of any computerised health information system. Without them other modern technologies such as decision support systems cannot be effectively integrated into routine clinical workflow. The paperless, interoperable, multi-provider, multi-specialty, multi-discipline computerised medical record, which has been a goal for many researchers, healthcare professionals, administrators and politicians for the past 20+ years, is however about to become reality in many western countries.
- In Kenya, the Division of Health Information Systems (HIS) has recognized the need to improve the use of ICT in health. Under Strategic Objective Five of the HIS Strategic Plan 2009-2014, the HIS aims to strengthen the use and application of information and communication technology, in data management. Tethered to this objective is the need to have standardized and interoperable ICT Applications, including Electronic Medical Records. It is with this objective in mind that the Ministries of Health, through the HIS, embarked on a process of standardization of EMRs in Kenya. Electronic Medical Record (EMR) systems are increasingly being adopted in Kenya to improve medical record management, health program management, and the quality of patient care.
- **WHAT IS AN EMR?**

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- EMR is an acronym for Electronic Medical Records. This refers to a paperless, digital and computerized system of maintaining patient data, designed to increase the efficiency and reduce documentation errors by streamlining the process. It can also be defined as a longitudinal collection of electronic health information that provides immediate, authorized access to person and population level data to support efficient health services delivery process.

CURRENT SITUATION ON THE USE OF ELECTRONIC MEDICAL RECORDS IN KENYA TODAY.

- Implementing EMR is a complex, expensive investment that has created a demand for Healthcare IT professionals and accounts for a growing segment of the healthcare workforce. Just like any other record keeping, moving patients' records from paper and physical filing systems to computers and their super storage capabilities creates great efficiencies for patients and their health care providers, as well as health payment systems
- In 2001 the Department of Medicine and Child Health and Paediatrics at Moi University, Eldoret and the department of internal Medicine and Geriatrics at the Indiana University school of Medicine in collaboration with the Moi Teaching and referral Hospital in Eldoret, Kenya, established the academic Model for Prevention and treatment of HIV/AIDS (AMPATH). Its medical records in Sub Saharan Africa are the first electronic medical record system for the Comprehensive management of the clinical care of patients infected with HIV. This system is composed of both paper-based and electronic records, has led to uniformity in data collection and facilitated the retrieval of patient data for clinical care and research.
- Other health facilities in the region both GOK and Private that have implemented the EMR in Kenya include Aga Khan Teaching and Referral Hospital, The Nairobi Hospital, Kisii Level 5 Hospital, MP Shah Hospital, Jaramogi Oginga Odinga Teaching and Referral Hospital, Coast Provincial general Hospital, Naivasha District Hospital among others.
- It has been observed that facilities that have embraced EMR have shown remarkable improvement in revenue collection as every activity is accounted for. It has also tools for stock control and supply chain management. This results in efficient use of resources as

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health care providers become more accountable and audit trail is possible i.e every action done in the system can be traceable to the real user.

- Some of the notable challenges with the EMR may include high cost of setting the infrastructure and maintenance of the system, lack of standardization of EMR in Kenya, computer viruses, and literacy level among users. Due to fear, change is considered a threat yet without adequate involvement of the users of the system, the project can never be developed.

ADVANTAGES AND DISADVANTAGES OF USING ELECTRONIC MEDICAL RECORDS.

- By making it easier to use and share information, EMRs can help health care providers do a better job of managing patient care. When fully functional and exchangeable, the benefits of EMRs offer far more than a paper record can. Some of the benefits include the ones discussed below:
- Money is saved by using electronic medical records; not just the cost of paper and file folders, but the cost of labor and space, too. In any business, time equals money. The efficiencies created by simply typing a few identifying keystrokes to retrieve a patient's record as opposed to staring at thousands of file folders, filing and refilling them -- saves a doctor's practice or a hospital many thousands of shillings. That's even taking the cost of the electronic system into account.
- Another benefit is safety. In the past, the way a doctor obtained your health history was by asking you. Each time you visited a new doctor's office, you filled out forms about your history, including previous surgeries, or the drugs you take on a regular basis. If you forgot a piece of information, or if you didn't write it down because it seemed unimportant to you, then your doctor didn't have that piece of your medical puzzle to work with.
- However, when doctors share records electronically, your new doctor only needs to ask your name, birth date, and possibly another piece of identifying information. She can then pull up your records from their electronic storage space. All of the information he needs to see will be there in full. When it comes time to diagnose you, it might be important to him to learn that you are taking a certain kind of medication, or even an herbal supplement -- any information shared with a previous doctor. Diagnosis and treatment

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decisions might be altered based on that information, which is far more complete than what you might have written down on paper.

- In the past, when a doctor closed his practice, retired, moved, or even died, patient records could easily get lost or relocated, making it impossible for patients to get the records they needed to take to a new doctor. Keeping these records electronically, especially in the cases where patients can also gain access to them, means the patient won't be left without the records she may need.
- EMRs save space. Instead of keeping huge paper files on patients, all records are kept on computer files. Though someone must store these records in computers, this still represents a small percentage of the space required to store physical records.
- Electronic medical records may save time as well. Though faxing and email assisted one doctor to get information from another doctor or a laboratory, there was generally a wait time. When a doctor has instant access to all of a patient's information, including things like x-rays, lab tests, and information about prescriptions or allergies, he or she is empowered to act right away, thus saving time.
- Many doctors are often considered to have undecipherable handwriting, and though this is a generalization, unclear writing can lead to mistakes. Typed information is less likely to create misunderstandings.
- One concern about the use of electronic medical records is that doctors may have a significant learning curve when these programs are first employed. A poor typist may actually take a long time to input information. Doctors often have to be their own medical clerks especially during an office visit, and a doctor distracted by confusing technology may not be as alert to a patient's symptoms or needs.
- There is no single electronic medical records source or system, so different hospitals and individual clinicians are not all using the same program. This negates the possibility of instant information for all on the medical team, since one program may not mesh with another.
- Some patients express concern that electronic medical records might be hacked and exploited by others. Misuse of private medical information could create problems for people who have conditions they wish to keep private.

CONCLUSION

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- Despite these concerns, it appears many doctors and hospitals are now attempting to use electronic medical records. It remains unclear how long it will take for old files with long medical histories to be updated into electronic means. It also seems that employing electronic means to store data still requires some thought so that information and systems are uniform.
- The Government of Kenya has come up with the national guidelines and standards for the implementation of Electronic Medical Records which will spearhead the growth in the country. Working with the implementing partners, health facilities like Rwambwa Health Centre in Siaya, Awendo Sub-District Hospital in Migori have embraced the EMR and effective and efficient management of health care system has been realized in these facilities.

GIS

What is a GIS?

Definition:

- It is a computer system capable of capturing, storing analyzing and displaying geographically referenced information i.e. data identified according to location.
- Information system designed to work with data referenced by spatial coordinates collected through the Global positioning system(GPS)

Components of a GIS

- **Hardware :** Computer, Network, printer, Plotter, digitiser
- **Software :** Database management system, statistical analysis system, map digitising system, geographic analysis system
- **Data :** The data that is input into the system in various forms such as maps, reports e.t.c
- **People :** People working with a GIS such as data entry staff, surveyors, IT administrators e.t.c
- **Processes/Methods :**Includes processes such as overlaying, digital mapping ,visual displaying, interpolation e.t.c

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Why use GIS?

- Ability to assimilate divergent sources of data
- both spatial and non-spatial
- Visualisation impact-what is where
- Analytical capability e.g. showing trends
- Overlaying of data(use of different layers)
- Modelling-What if? Sharing of information

Remind participants about the John Snow experiment where John snow by use of geographic coordinates was able map out the houses of people that had cholera and through this he was able to isolate the circumstances that seemed to favour the spread of cholera, key among them being poor sanitation and refuse disposal.

The importance of GIS in healthcare: -

- GIS helps in visualising data e.g. in surveillance you are able to locate and monitor trends.
- Helps to identify the areas of greatest health needs to enable decision makers to prioritise resources.
- Also one may map conditions before and after action or event to see the impact. A SCHMT might map the change in the spread of a disease before and after an epidemic.
- Modelling enables one to see **what if** situations in comparison to new features in a location e.g. new road, new market, health facility e.t.c.
- We can also use different layers of presentation such as geographic, health facility coverage, population etc. and this can aid in visualisation for decision making

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The role of computers in HIS

Review relevant HIS software

Computer application

- Preparation of data entry sheets

Procedure of computer data in-put and out-put

Various in-put media

Computer operation

Validation of data

Data processing

- Analysis
- Data interpretation
- Information out-put from the computer
- Information dissemination levels and users

Data presentation:-

Purpose of data presentation

- Identify patterns and trends within the data
- Summarize results of data analysis
- Communicate information to others
  - Periodic data summaries
  - Investigation reports
  - Slide presentations and posters
  - Scientific publications

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Choosing the most appropriate format for displaying data depends on:

- Variable type
- Purpose of the data presentation
- Complexity of the data
- Ease of interpretation
- Effectiveness of conveying your message

Data is presented differently depending on variable type

1. **Qualitative or Categorical variables** cannot be measured numerically. Instead, they are divided into non-numerical categories. Examples of qualitative variables include hair color, occupation, region.

Qualitative variables are further divided into 2 categories:

(a) **Nominal variables** are those whose categories are labels or names that don't have an inherent order. For example, Nationality – Ugandan, Kenyan, and Tanzanian.

(b) **Ordinal variables** are those whose categories are non-numerical, but still have some order to them. Think of the word "order" to help you remember. For example, Severity of illness—mild, moderate, severe

2. **Quantitative variable**. Think of the word quantity. Quantitative variables can be measured numerically.

(a) Discrete variable. These variables are measured as whole numbers. For example, Count Data – 1, 2, 3, 4… n such as number of cups of tea consumed.

(b) Continuous variable. These variables can have numerical measurements in between whole numbers. For example, height can be 175.15 cm.

Data can be presented in

- Tables
- Bar Charts
- Histograms

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- Line Graphs
- Scatter plots
- Maps
- Pie Charts
- Box plots
- Frequency Polygons
- Survival Curves

- Tables, charts, and maps are common ways to display data
- Presented data should be clear, meaningful, and easy to interpret
- Effective visual displays avoid unnecessary or distracting features

Administrative and routine

Contraceptive prevalence rate (CPR)

Definition:

The percent of women of reproductive age who are using (or whose partner is using) a contraceptive method at a particular point in time, almost always reported for women married or in sexual union.

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Generally, the measure includes all contraceptive methods (modern and traditional), but it may include modern methods only.

The indicator is calculated as follows:

(Number of women 15-49 using a contraceptive method / total number of women 15-49) x 100

Illustrative Example

The DHS for Peru (2000) yielded the following data on CPR, among women 15-45 years of age:

All women	**Women currently married or in union**
CPR= 12,240/27,843	CPR = 10,764/15,628
=0.440 x 100	= 0.689 x 100
=44.0	= 68.9

Data Requirement(s):

The total number of women of reproductive age, by marital status; and of these, the number that are currently using a contraceptive method

Data Source(s):

Population-based surveys

Purpose:

The CPR provides a measure of population coverage of contraceptive use, taking into account all sources of supply and all contraceptive methods; it is the most widely reported measure of outcome for family planning programs at the population level.

Technically speaking, CPR is a ratio, not a rate. (Prevalence is measured by a ratio and incidence by a rate.) For a given year, contraceptive prevalence measures the percentage of women of childbearing age in union who use a form of contraception. To obtain a true contraceptive use rate, the denominator should reflect the population at risk (of pregnancy), i.e., sexually active women who are not infecund, pregnant, or amenhorreic. The numerator should reflect the number of contraceptive users from that population. (Note: We include this point for

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informational purposes only.) The international population community uses the term "contraceptive prevalence rate" as defined above; thus, this database endorses this practice to assure consistency.

Issue(s):

The convention in reporting contraceptive prevalence is to base this calculation on women married or in sexual union (even though most DHS-type surveys ask questions of contraceptive use to women of reproductive age, regardless of their marital status). In countries with relatively little sexual activity outside marriage for women, basing prevalence estimates on women in sexual union captures the population at risk of pregnancy. However, in countries with the widespread practice of sexual activity outside of marriage or stable sexual unions, a prevalence estimate based on women in union *only* would ignore a considerable percentage of current users. Thus, researchers and program evaluators generally report percentage of sexually active unmarried women using contraception, if appropriate, in addition to contraceptive prevalence, because method mix is very different for those married versus unmarried (in/not in a stable union).

Whereas evaluators may theoretically derive the CPR from service statistics on numbers of current users and estimates of the population at risk, current practice is to rely upon population-based sample surveys in order to minimize the problems associated with maintaining a running count of current users and with obtaining accurate population estimates. (The problems include incomplete data, double-counting of users who enter the service delivery system at more than one point, purposeful inflation of service statistics, and poor quality of data due to other activities competing for the attention of those recording the information, to name the primary ones.)

The DHS and RHS are currently the main sources for obtaining national level estimates of prevalence. ("DHS" is used in this database to mean "DHS-type surveys:" the actual DHS, the RHS surveys conducted with technical support from CDC, and other large-scale national surveys conducted by the countries themselves under other auspices). Evaluators may also use smaller scale and/or more focused surveys to estimate the CPR as long as they use probability sampling methods, the essential ingredient for obtaining scientifically sound estimates. Evaluators may also obtain CPR by adding relevant questions to surveys on other topics (e.g., health program prevalence or coverage surveys), assuming appropriate sampling methods and sample sizes.

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Keywords:

long-acting/permanent methods (LAPM), family planning, safe motherhood (SM)

Poverty and Equity Considerations:

(Excerpted from: Becker L, Wolf J, Levine R (2006) *Measuring commitment to health*. Center for Global Development.)

A study of public family planning service usage found that use s from the wealthiest quintile outnumbered those from the poorest quintile in 13 of the 20 developing countries examined, and that the contraceptive prevalence rate is significantly higher amongst the wealthiest quintile in all 20 countries. However, the study also found that countries with a higher CPR had less disparity than those in which a smaller percentage used contraceptives, indicating that increasing the CPR could contribute to reducing inequity

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